# DRL-Based HVAC Control for a Single-Zone Smart Building: A Comparative Study of Control Agents

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***Abstract*—Buildings account for a large share of global energy use, and HVAC is one of their largest controllable loads. This paper studies deep reinforcement learning for continuous HVAC control in a single-zone smart building and compares PPO, A2C, SAC, and TD3 against a tuned PID baseline. The environment combines nonlinear heat transfer, a Carnot-based coefficient of performance, monthly electricity tariffs, domain randomization, and balanced seasonal sampling. Agents are trained on Syracuse weather and evaluated on held-out Albany weather to assess geographic generalization. Over 100 evaluation episodes, TD3 achieves the best median performance with $0.12\,^{\circ}\mathrm{C}$ MAE, $16.9\,\mathrm{kWh/day}$ daily energy use, and $99.7\%$ of time within $\pm 0.5\,^{\circ}\mathrm{C}$ of the setpoint. A2C and SAC also generalize, while PPO trails the other DRL agents. All four DRL methods outperform PID, which achieves $1.01\,^{\circ}\mathrm{C}$ median MAE and $63.2\%$ comfort. A lightweight simulator paired with seasonal diversity and parameter uncertainty during training supports transfer to unseen weather.**

***Index Terms*—Deep reinforcement learning, HVAC control, smart buildings, energy efficiency, PPO, SAC, TD3, A2C, PID, domain randomization**

## I. Introduction

Buildings are among the largest energy consumers worldwide, and heating, ventilation, and air conditioning (HVAC) is a major share of that demand. HVAC control is hard because outdoor temperature, solar gains, wind speed, and electricity prices vary over time, and building envelopes differ across sites.

Traditional strategies include rule-based control, proportional integral derivative (PID) control, and model predictive control (MPC). MPC plans ahead but depends on accurate models, calibrated parameters, and reliable forecasts, which makes deployment costly. deep reinforcement learning (DRL) learns a policy that maps observations directly to actions, without solving an optimization problem at every timestep. The open question is generalization: a controller trained in one climate or simulator may degrade when deployment conditions change.

We study this problem in a compact setting. A single-zone thermal model with nonlinear conductance, solar gains, a coefficient of performance (COP)-aware energy term, and monthly electricity prices is used to compare four continuous-control DRL agents (PPO, A2C, SAC, TD3) against a tuned PID baseline. Training uses Syracuse weather only and evaluation uses Albany weather only, giving a geographic out-of-distribution (OOD) split. Training also uses domain randomization over thermal parameters and round-robin seasonal sampling.

The contributions are: (i) a controlled comparison of four DRL agents and PID under the same task, observation, action, and reward; (ii) generalization measured by a city-level train/test split; and (iii) evidence that a lightweight simulator with domain randomization and balanced seasonal exposure produces policies that transfer to unseen weather.

## II. Related Work

Recent surveys report that deep reinforcement learning (DRL) can match or exceed MPC on comfort-energy trade-offs while raising open issues in reward design, reproducibility, and sensitivity to changing conditions [1]–[3].

For continuous control, A2C/A3C introduced asynchronous advantage estimation [4]. PPO replaced trust-region updates with a clipped surrogate objective and is now a common on-policy baseline [5]. TD3 reduced value overestimation through clipped double critics, delayed policy updates, and target policy smoothing [6]. SAC added maximum-entropy learning with a stochastic actor and good sample efficiency [7].

The Sinergym benchmark [1] evaluated PPO, TD3, and SAC across buildings and climates and found that off-policy methods often produce better trade-offs, while noting that many HVAC papers compare only one or two algorithms in a single setting. On the transfer side, a continual DRL study [2] reported that transfer and sample efficiency remain bottlenecks, and a model-based SAC framework with hypernetworks improved transfer and reduced catastrophic forgetting. Coupling DRL HVAC control with an urban climate model [3] showed that background climate shifts both the reward trade-off and policy transferability across cities.

Our work is closest to [1] but targets a different gap: a simpler single-zone setting that isolates algorithmic differences, a strict city-level transfer split, and a cost-aware reward that depends on both electricity price and instantaneous COP.

## III. Environment Design

Figure 1 summarizes the full workflow. Raw weather and tariff data are parsed into hourly exogenous inputs, which drive

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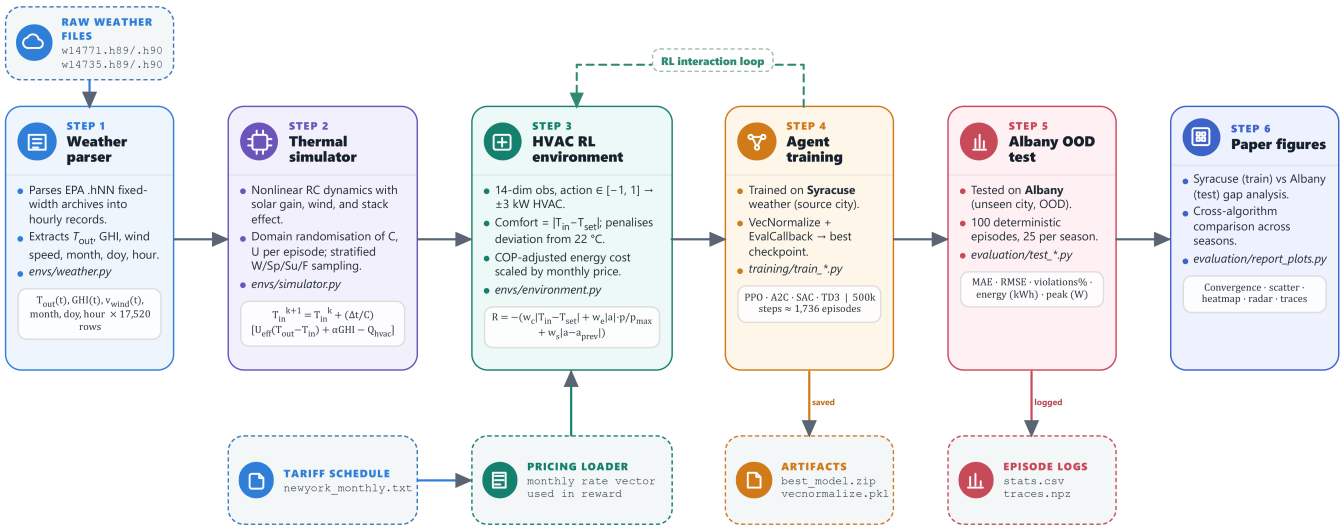



Fig. 1. End-to-end workflow. EPA weather archives and NY electricity tariffs drive a nonlinear lumped-capacitance simulator with domain randomization and stratified seasonal sampling, exposed as a 14-dimensional Gymnasium environment. PPO, A2C, SAC, and TD3 are trained for 500 k steps on Syracuse and evaluated out-of-distribution on Albany over 100 episodes (25 per season).

a five-minute physics-based simulator exposed as a Gymnasium environment. This section describes the data pipeline, the thermal model, and the episode design choices used to improve generalization.

### A. Weather and Tariff Inputs

We use U.S. Environmental Protection Agency (EPA) PRZM-format hourly weather archives from 1989 to 1990 for two upstate New York stations: Syracuse (station 14771), used for training only, and Albany (station 14735), used for evaluation only. Each station contributes about 17 520 hourly records of outdoor temperature, global horizontal irradiance (GHI), and wind speed. The split is geographic and enforced at the file-path level in the codebase, so Albany is never used during training, hyperparameter tuning, or model selection.

The PRZM `.hNN` files are fixed-width ASCII tables. A custom parser extracts the date (cols. 2–11, `yyyy-mm-dd`), the EPA hour index (cols. 12–14, where 1 maps to 00:00 and 25 is a daily summary row that is discarded), global horizontal irradiance (GHI) in $\mathrm{W\,h\,m^{-2}}$ (cols. 30–34), dry-bulb outdoor temperature in $^\circ\mathrm{C}$ (cols. 65–69), and wind speed at $10\,\mathrm{m}$ in $\mathrm{m\,s^{-1}}$ (cols. 96–100). Trailing quality-flag characters are stripped. Missing values, encoded as `---`, are forward-filled and then back-filled to remove short gaps. Small negative global horizontal irradiance (GHI) and wind values caused by rounding artifacts are clipped to zero. The annual files for each station are then concatenated and sorted into a single hourly DataFrame with columns `T_out`, `GHI`, `wind_speed`, `month`, `doy`, and `hour_of_day`.

Electricity prices are taken from the NYSERDA monthly average residential retail rate table for New York State. The loader extracts a 12-element NumPy vector $p \in \mathbb{R}^{12}$ in cents per $\mathrm{kW\,h}$, indexed by calendar month. During simulation, each five-minute control step uses a zero-order hold on the most recent hourly weather row, so the same $T_{\text{out}}$, global horizontal irradiance (GHI), and $v_{\text{wind}}$ values are applied for twelve consecutive simulator steps. The month field is used to look up $p_t = p[\text{month} - 1]$, which is exposed to the agent in normalized form and also used in the reward.

TABLE I
THERMAL SIMULATOR PARAMETERS

| Symbol | Description | Value / Range |
|---|---|---|
| $C$ | Thermal capacitance | $\mathcal{U}(200\,\mathrm{k},\ 500\,\mathrm{k})\ \mathrm{J/^\circ C}$ |
| $U_{\text{cond}}$ | Conduction coefficient | $\mathcal{U}(30,\ 80)\ \mathrm{W/^\circ C}$ |
| $k_w$ | Wind infiltration coeff. | $4.45\ \mathrm{W/^\circ C}$ |
| $n_w$ | Wind power-law exponent | 0.65 |
| $k_s$ | Stack-effect coefficient | $2.6\ \mathrm{W/^\circ C}$ |
| $\alpha$ | Solar gain coefficient | $0.5\ \mathrm{m^2}$ |
| $\eta_{\text{COP}}$ | Carnot efficiency factor | 0.4 |
| $Q_{\max}$ | Maximum HVAC power | $3000\ \mathrm{W}$ |
| $T_{\text{set}}$ | Temperature setpoint | $22\ ^\circ\mathrm{C}$ |
| $\Delta t$ | Control timestep | $300\ \mathrm{s}$ |

### B. Thermal Zone Dynamics

The simulator follows a first-law lumped-capacitance model for a single-zone building, integrated with a forward Euler step:

$$T_{\text{in}}^{k+1} = T_{\text{in}}^{k} + \frac{\Delta t}{C}\left[U_{\text{eff}}\big(T_{\text{out}}^{k} - T_{\text{in}}^{k}\big) + \alpha \cdot \text{GHI}^{k} - Q_{\text{hvac}}^{k}\right] \quad (1)$$

where the effective envelope conductance combines conductive transfer, wind-driven infiltration, and stack effects:

$$U_{\text{eff}} = U_{\text{cond}} + k_w\, v_{\text{wind}}^{n_w} + k_s\sqrt{|T_{\text{in}} - T_{\text{out}}|} \quad (2)$$

Table I lists the physical parameters and the randomized ranges used during training. The simulator also computes an instantaneous Carnot-based coefficient of performance (COP),

TABLE II
OBSERVATION SPACE (14-DIMENSIONAL STATE VECTOR).

| # | Feature | Range | Group | Description |
|---|---|---|---|---|
| 0 | $T_{\text{in}} - T_{\text{set}}$ | $[-50, 50]\,^{\circ}\text{C}$ | Thermal | Indoor temperature error |
| 1 | $(T_{\text{out}} - 20)/20$ | $[-2, 2]$ | Thermal | Outdoor temperature, centered and scaled |
| 2 | $\sin(2\pi h/24)$ | $[-1, 1]$ | Time | Hour sine encoding |
| 3 | $\cos(2\pi h/24)$ | $[-1, 1]$ | Time | Hour cosine encoding |
| 4 | $\sin(2\pi d/365)$ | $[-1, 1]$ | Time | Day sine encoding |
| 5 | $\cos(2\pi d/365)$ | $[-1, 1]$ | Time | Day cosine encoding |
| 6 | $p_t/p_{\max}$ | $[0, 1]$ | Exogenous | Price normalized by annual maximum |
| 7 | $\text{GHI}/1200$ | $[0, 1]$ | Exogenous | Solar irradiance normalized by 1200 |
| 8 | $v_{\text{wind}}/20$ | $[0, 1]$ | Exogenous | Wind speed normalized by 20 |
| 9 | $C/C_{\text{nom}}$ | $[0.57, 1.43]$ | DR | Capacitance ratio |
| 10 | $U_{\text{cond}}/U_{\text{cond,nom}}$ | $[0.55, 1.45]$ | DR | Conduction-coefficient ratio |
| 11 | $a_{t-1}$ | $[-1, 1]$ | Actuator | Previous action |
| 12 | $\text{COP}/10$ | $[0.08, 1]$ | Physics | COP normalized by 10 |
| 13 | $U_{\text{eff}}/150$ | $[0, 1]$ | Physics | Effective conductance normalized by 150 |

which appears in both the observation vector and the reward function.

### C. Episode Design for Generalization

Two mechanisms improve generalization during training. First, at each episode reset, the simulator randomizes the thermal capacitance $C$ and conduction coefficient $U_{\text{cond}}$ over the ranges in Table I, spanning light to heavy constructions and well-insulated to leaky envelopes. The agent receives the normalized ratios $C/C_{\text{nom}}$ and $U_{\text{cond}}/U_{\text{cond,nom}}$ so it can adapt online to the current envelope.

Second, episode start indices are sampled in a season-balanced way. Random sampling from a two-year archive can overrepresent mild periods, so valid 24-hour start indices are partitioned into winter, spring, summer, and fall buckets. At each reset, the simulator cycles through these buckets in round-robin order and draws a random start index from the active season. The same scheme is used during evaluation: the Albany test set contains 100 episodes, with 25 from each season.

## IV. REINFORCEMENT LEARNING FORMULATION

We formulate HVAC control as a continuous-state, continuous-action Markov decision process (MDP). At each timestep, the agent observes the current thermal and exogenous conditions, chooses an HVAC command, and receives a reward that trades off comfort, energy cost, and actuator smoothness.

### A. Observation Space

The observation is a 14-dimensional continuous vector summarized in Table II. Features are grouped into thermal state, time encodings, exogenous inputs, randomized building parameters, and physics-aware variables. Cyclic sine/cosine features encode hour of day and day of year without discontinuities. Exposing the randomized ratios $C/C_{\text{nom}}$ and $U_{\text{cond}}/U_{\text{cond,nom}}$ lets the policy adapt to different envelopes inside the randomized simulator.

### B. Action Space

The agent outputs a single continuous action $a_t \in [-1, 1]$, mapped linearly to HVAC power through $Q_{\text{hvac}} = a_t Q_{\max}$. Positive $a_t$ commands cooling, negative $a_t$ commands heating, and $a_t = 0$ turns the HVAC off. The continuous formulation lets the controller modulate power smoothly instead of switching between discrete operating modes.

### C. Reward Function

The reward balances temperature tracking, energy cost, and control smoothness:

$$R_t = -\Big(\underbrace{w_c\,|T_{\text{in}} - T_{\text{set}}|}_{\text{comfort}} + \underbrace{w_e\,\frac{|a_t|}{\text{COP}_t}\cdot\frac{p_t}{p_{\max}}}_{\text{energy cost}} + \underbrace{w_s\,|a_t - a_{t-1}|}_{\text{slew penalty}}\Big) \tag{3}$$

The first term penalizes deviation from the setpoint. The second term scales action magnitude by the instantaneous coefficient of performance (COP) and the current electricity price, so the same actuation is more expensive when the system is less efficient or electricity is costlier. The third term discourages abrupt changes in HVAC power. Unless stated otherwise, the weights are $w_c = 1.0$, $w_e = 0.1$, and $w_s = 0.05$, making comfort the primary objective while keeping the controller cost-aware and smooth.

## V. CONTROLLERS AND EXPERIMENTAL SETUP

### A. Controllers Compared

We compare four continuous-control DRL algorithms implemented in Stable-Baselines3 (SB3) [8]: PPO [5], A2C [4], SAC [7], and TD3 [6]. A tuned PID controller is included as a conventional baseline.

*a) PPO.:* PPO is an on-policy policy-gradient method that updates the actor with a clipped surrogate objective. The clipping limits how far the new policy can move from the old one in a single update, which improves stability. PPO is a common baseline, but its on-policy nature makes it less sample-efficient than off-policy methods.

TABLE III
TRAINING HYPERPARAMETERS

| Parameter | PPO | A2C | SAC | TD3 |
|---|---|---|---|---|
| Total steps | | 500 000 | | |
| Policy | | MlpPolicy [64, 64] | | |
| Seed | | 42 | | |
| # envs | 8 | 8 | 1 | 1 |
| Learning rate $\alpha$ | 3e-4 | 7e-4 | 3e-4 | 3e-4 |
| Discount factor $\gamma$ | 0.98 | 0.99 | 0.98 | 0.98 |
| Batch size | 64 | – | 256 | 256 |
| $n_{\text{steps}}$ | 2048 | 5 | – | – |
| Buffer size | – | – | 1 M | 1 M |
| Target update rate $\tau$ | – | – | 0.02 | 0.005 |
| Entropy coeff. | 0.01 | 0.01 | auto | – |
| Action noise | – | – | – | $\mathcal{N}(0, 0.1)$ |
| Reward normalization | yes | yes | no | no |

*b) A2C.:* A2C belongs to the actor-critic family introduced by Mnih et al. [4]. It learns a policy and a value baseline jointly, using advantage estimates to reduce gradient variance. Compared with PPO, A2C is lighter and often faster to train, but can be more sensitive to rollout settings.

*c) SAC.:* SAC is an off-policy actor-critic method based on a maximum-entropy objective. It learns a stochastic policy that seeks high reward while preserving entropy for exploration. Replay-buffer reuse and entropy regularization fit HVAC control, where data efficiency and stable learning matter.

*d) TD3.:* TD3 is an off-policy deterministic actor-critic method designed to reduce value overestimation in continuous control. It uses two critics, delayed policy updates, and target-policy smoothing. These choices fit precise HVAC actuation, where small value errors can lead to poor power commands.

*e) PID baseline.:* A proportional integral derivative (PID) controller is included to contextualize the DRL results. It computes HVAC power from the temperature error $e_t = T_{\text{set}} - T_{\text{in}}$ using fixed gains tuned on the Syracuse training environment. Unlike the DRL agents, PID does not observe weather variables, electricity price, or randomized building parameters, and its gains do not adapt across episodes.

### B. Training Configuration

All DRL agents are trained on Syracuse weather only, with hyperparameters listed in Table III. PPO and A2C are on-policy and require fresh rollouts, so they use eight parallel `SubprocVecEnv` workers. SAC and TD3 are off-policy and learn from replayed transitions, so a single `DummyVecEnv` suffices. Observations are normalized using `VecNormalize`.

### C. Evaluation Protocol

After training on Syracuse, each learned policy is frozen and transferred to Albany, the geographic out-of-distribution (OOD) test set. No Albany weather data are used during training, tuning, or model selection. Evaluation is deterministic: the controller is applied without exploration noise or action sampling.

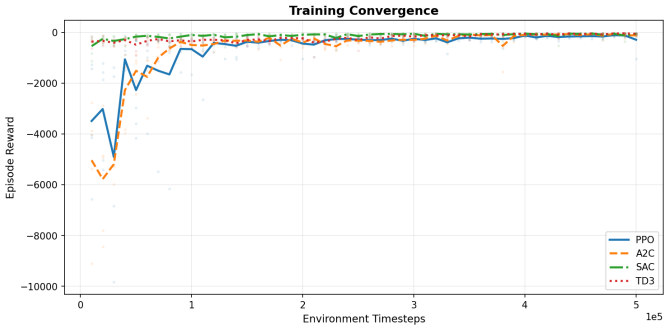



Fig. 2. Mean evaluation reward measured at regular checkpoints during 500 000-step training on Syracuse, with interquartile-range shading.

Each evaluation episode is one closed-loop simulation rollout, and 100 Albany episodes are used, with 25 drawn from each season. Per episode we record four physical metrics and report the median: mean absolute error (MAE) and root mean square error (RMSE) of indoor temperature relative to the setpoint, the percentage of time within $\pm 0.5\,^{\circ}\mathrm{C}$ of the setpoint, and daily HVAC energy consumption. Reward is used only during training, and final comparison is based on the physical metrics. Median values reduce sensitivity to unusually difficult episodes.

## VI. RESULTS AND DISCUSSION

### A. Learning Curves

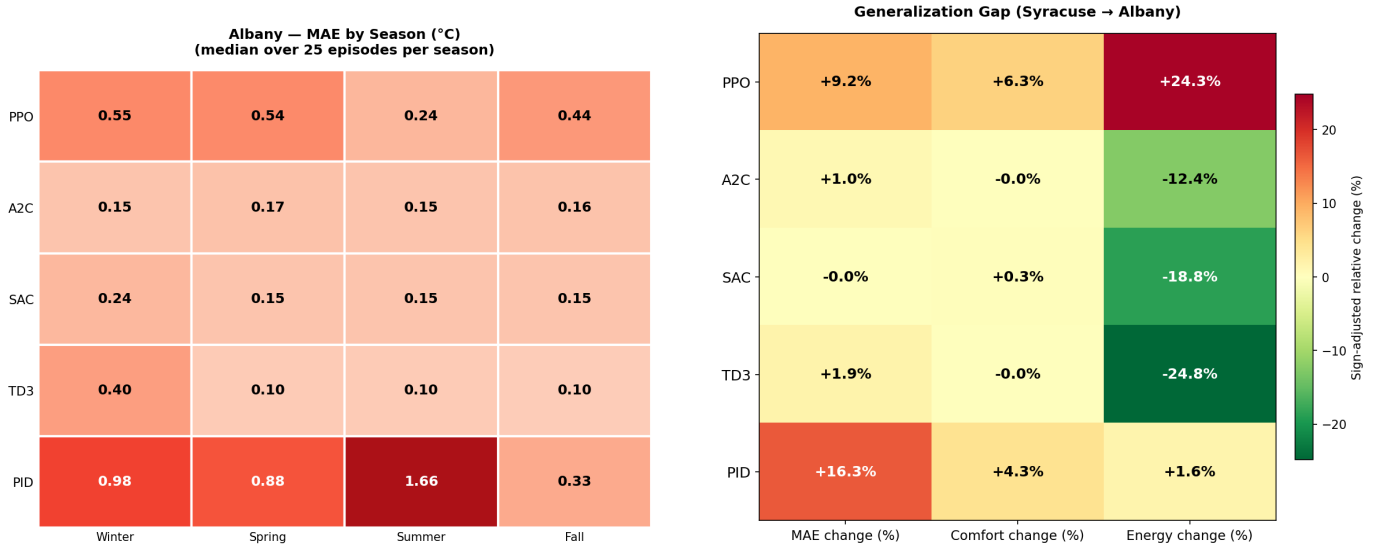
Figure 2 shows the mean evaluation reward at regular checkpoints during training on Syracuse. A reward of 0 corresponds to a comfortable, energy-free reference, so the negative values arise from the three penalty terms in (3). All four algorithms reach a stable regime well before the full training budget is exhausted.

SAC and TD3 converge faster and to a higher plateau than the on-policy methods. Both reuse transitions through a replay buffer, which improves sample efficiency, and SAC benefits from entropy regularization during early exploration. TD3 gains stability from its twin-critic architecture, which reduces Q-value overestimation. Both off-policy methods approach a near-zero reward plateau by roughly 150 000–200 000 steps. A2C is competitive despite greater early variance, while PPO converges more conservatively and to a lower final plateau.

### B. Albany Out-of-Distribution Performance

Under the default joint $C{+}U_{\text{cond}}$ randomization, A2C, SAC, and TD3 transfer to the unseen Albany climate without retraining. TD3 is the best controller, reaching $0.12\,^{\circ}\mathrm{C}$ MAE and $99.7\%$ comfort, while A2C and SAC follow closely. PPO outperforms PID but trails the other DRL agents, with $0.45\,^{\circ}\mathrm{C}$ MAE and $64.9\%$ comfort. All four DRL methods beat PID, which achieves $1.01\,^{\circ}\mathrm{C}$ MAE, $63.2\%$ comfort, and $42.7\,\mathrm{kWh/day}$ energy use.

Winter is the hardest season for all agents because the larger indoor-outdoor temperature gap increases tracking error and heating demand. Figure 3a shows the per-season MAE on




(a) Per-season median MAE on Albany.

(b) Median train-to-test gap (Syracuse→Albany), where 0 means no change and positive values indicate worse Albany performance.

Fig. 3. Albany out-of-distribution performance. (a) Per-season median MAE (°C) over 25 episodes per season. (b) Percentage change in each metric when transferring from Syracuse to Albany (lower is better in both panels).

TABLE IV
DOMAIN RANDOMIZATION ABLATION ON ALBANY. BEST PER AGENT IN BOLD.

| Agent | Metric | No DR | Rand $C$ | Rand $U_{\text{cond}}$ | Rand $C$+$U_{\text{cond}}$ |
|---|---|---|---|---|---|
| PPO | MAE (°C) | 2.33 | 0.55 | 0.63 | **0.45** |
| | Comfort (%) | 13.95 | 51.43 | 45.43 | **64.89** |
| | Energy (kWh/day) | 24.74 | 27.24 | 25.51 | **21.71** |
| A2C | MAE (°C) | 1.99 | 2.97 | 1.69 | **0.16** |
| | Comfort (%) | 15.70 | 9.07 | 18.51 | **99.68** |
| | Energy (kWh/day) | 23.57 | 22.75 | 21.21 | **18.50** |
| SAC | MAE (°C) | 9.78 | 6.56 | 0.19 | **0.17** |
| | Comfort (%) | 1.17 | 0.69 | 98.06 | **98.96** |
| | Energy (kWh/day) | 40.62 | 28.90 | 24.51 | **21.03** |
| TD3 | MAE (°C) | 3.17 | 2.49 | 8.93 | **0.12** |
| | Comfort (%) | 3.23 | 18.17 | 25.81 | **99.70** |
| | Energy (kWh/day) | 28.13 | 25.41 | 44.38 | **16.90** |

PID baseline: MAE 1.01 °C, comfort 63.2%, 42.7 kWh/day.

Albany. A2C maintains near-perfect tracking across all four seasons, TD3 matches it in spring, summer, and fall, and PPO drops sharply outside the easiest conditions. The TD3 winter and summer traces in Figure 4 illustrate this: in winter, TD3 sustains high heating against very low outdoor temperatures while staying close to the comfort band; in summer, the controller uses smoother and lower HVAC effort. Figure 3b shows the percentage change in performance when transferring from Syracuse to Albany.

### C. Domain Randomization Ablation

To isolate the effect of parameter randomization, each agent is trained under four configurations: no DR, random $C$ only, random $U_{\text{cond}}$ only, and joint $C$+$U_{\text{cond}}$. Table IV shows that joint randomization is the best configuration across algorithms. For PPO, A2C, and TD3, low Albany error appears only when both parameters are randomized together. SAC benefits from $U_{\text{cond}}$ randomization alone but achieves its best Albany performance under joint randomization. Single-parameter randomization is inconsistent across agents, while joint randomization yields consistent transfer.

### D. Discussion

Five observations stand out. First, TD3 is the best method in this setting. Its deterministic policy fits a low-dimensional continuous-control task in which precise power levels matter, and its twin-critic design reduces value overestimation that could otherwise lead to poor actuator commands. Combined with target-policy smoothing, TD3 produces smooth HVAC traces (Figure 4) and the lowest median daily energy use of any controller.

Second, A2C is a stronger baseline than expected. Although simpler than SAC and TD3 and trained without a replay buffer, it reaches nearly the same comfort level and remains close on tracking error. This suggests the single-zone task is structured enough that a lightweight actor-critic with parallel rollouts can extract most of the available performance. The practical implication is that a heavy off-policy pipeline is not strictly required to obtain good HVAC control on this class of problems.

Third, PID is the worst controller across every metric. Compared with TD3, PID uses about 2.5× more energy while spending only about two-thirds as much time inside the comfort band, even after gain tuning on Syracuse. PID has no access to weather variables, electricity price, or the randomized envelope, and its fixed gains cannot account for thermal inertia or anticipate future conditions. The gap to

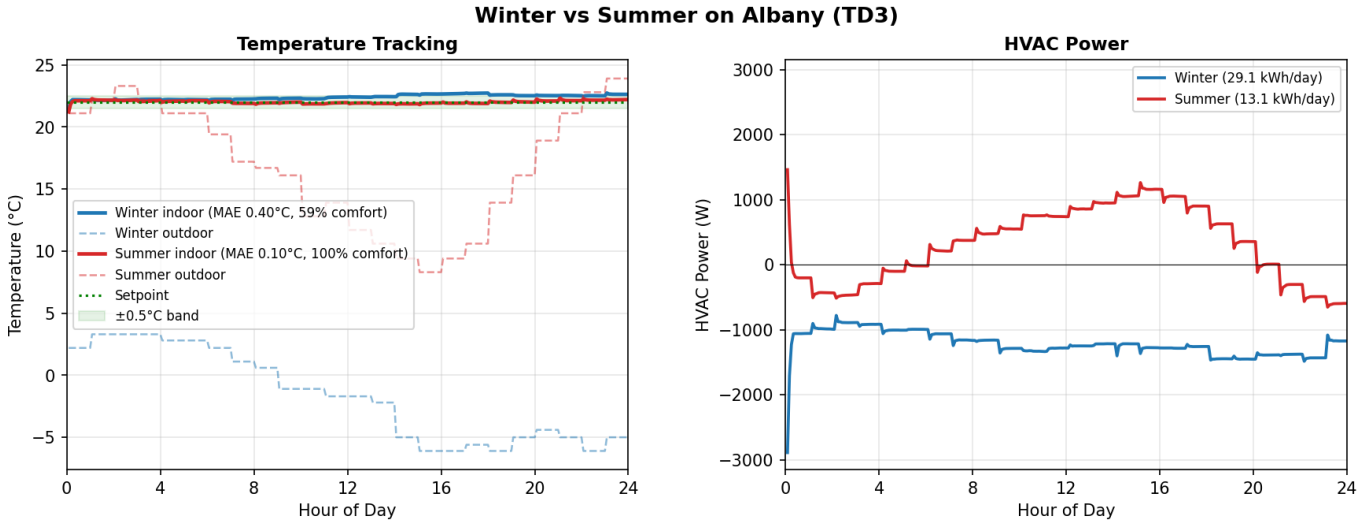



Fig. 4. Representative winter and summer Albany episodes for TD3. Left: indoor and outdoor temperature traces with the $\pm 0.5\,^{\circ}\mathrm{C}$ comfort band in green. Right: HVAC power profile for each season.

the DRL agents shows the value of state-aware, cost-aware policies that learn to balance comfort and price over a horizon rather than reacting to the instantaneous error.

Fourth, transfer is not explained by algorithm choice alone. The ablation in Table IV shows that single-parameter randomization can be misleading: TD3 actually *degrades* when only $U_{\text{cond}}$ is randomized (MAE rises to $8.93\,^{\circ}\mathrm{C}$), and PPO is worst with no randomization at all. Joint $C{+}U_{\text{cond}}$ randomization is the only setting in which all four agents reach their best Albany numbers, indicating that exposure to coupled envelope and capacitance variation is what drives generalization, not the algorithm family.

Fifth, PPO's larger transfer gap suggests that its on-policy update, combined with a clipped objective tuned for stability, does not exploit the 500 000-step budget as efficiently as the off-policy methods. A longer schedule, a smaller clip range, or a tuned learning-rate schedule could narrow the gap, but on the present budget PPO is clearly the most conservative of the four DRL agents.

The study also has limitations. The simulator is single-zone and ignores occupancy, internal gains, and humidity. The PID baseline is a fixed-gain controller, not an MPC, so the comparison establishes a floor rather than the state of the art among model-based methods. Evaluation also relies on a single train/test city pair, so the absolute transfer numbers should be read as evidence that geographic generalization is achievable rather than as a calibrated gap across climates.

## VII. Conclusion

This paper compared four DRL agents (PPO, A2C, SAC, and TD3) for continuous single-zone HVAC control under a common nonlinear simulator, reward function, and evaluation protocol. All agents were trained on Syracuse weather and evaluated on 100 deterministic Albany episodes, providing a geographic out-of-distribution test that is stricter than the usual same-city held-out split.

TD3 was the best controller, reaching $0.12\,^{\circ}\mathrm{C}$ median MAE and $99.7\%$ time inside the comfort band on Albany, with the lowest energy use of any agent. A2C and SAC matched it on comfort and tracking, while PPO, the weakest DRL agent, still outperformed PID on every metric. All four DRL agents reduced median daily energy use by 49–60% relative to PID while improving comfort, supporting the case that learned policies can trade off comfort and energy more effectively than a fixed-gain baseline.

The ablation in Table IV indicates that joint randomization of $C$ and $U_{\text{cond}}$ is the dominant driver of generalization: every agent reached its best Albany numbers under joint randomization, and single-parameter randomization was inconsistent or actively harmful. Combined with round-robin seasonal sampling, this exposed agents to coupled envelope and weather variation during training and avoided overfitting to mild operating periods. Together these results argue that a compact physics-informed simulator, paired with the right training-time exposure, is enough to obtain controllers that transfer to unseen weather without retraining.

Several directions remain open. Multi-zone buildings with inter-zone coupling, occupancy-driven setpoint scheduling, demand response events, and dynamic real-time tariffs would each stress the controller in ways not captured by the present setup. On the algorithmic side, a larger training budget, PPO-specific tuning, and adding a model predictive control baseline would tighten the comparison. Finally, evaluating transfer across more city pairs and climate zones would convert the present geographic split into a calibrated measure of how DRL HVAC controllers degrade as the deployment climate moves further from the training distribution.

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